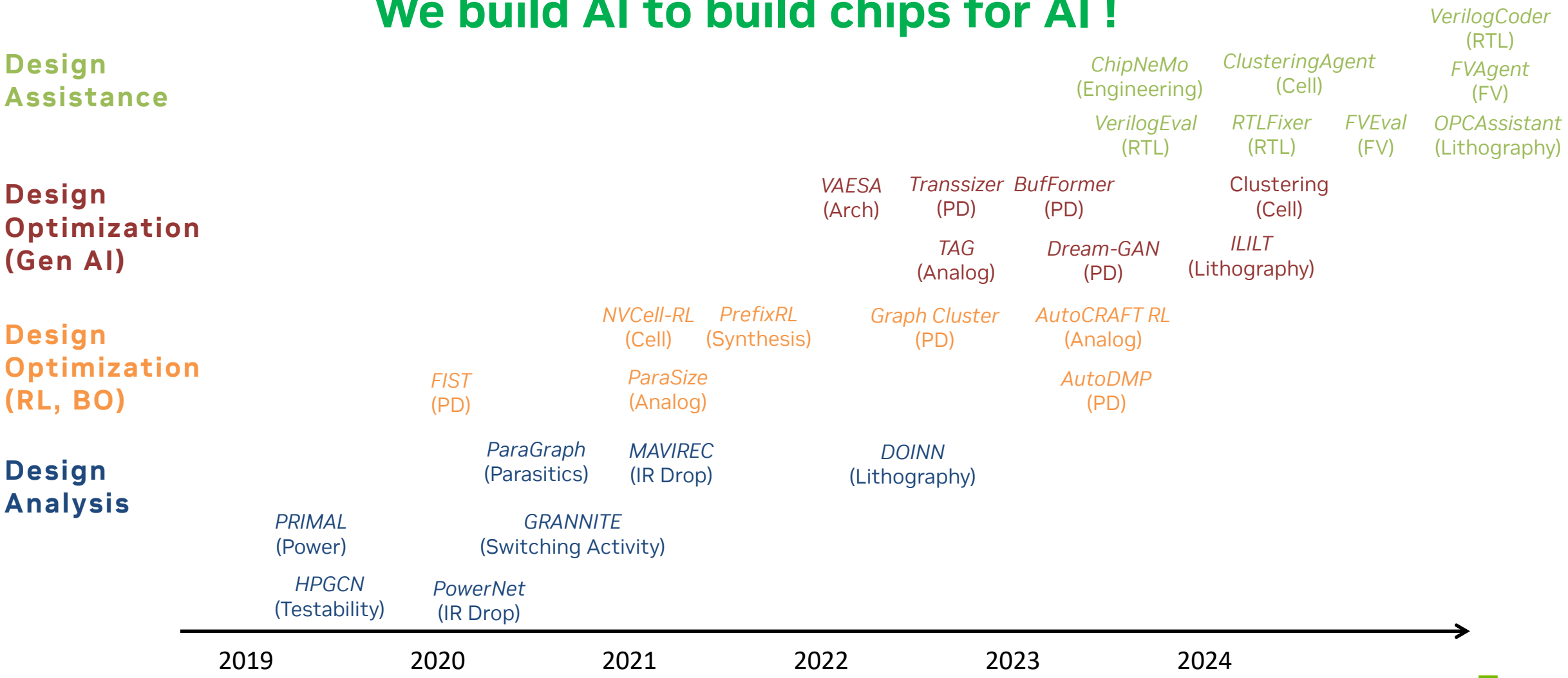
The Application of AI to Chip Design

Haoxing (Mark) Ren, Director of Design Automation Research, NVIDIA

AI for Chip Design Research @ NVIDIA

We build AI to build chips for AI !



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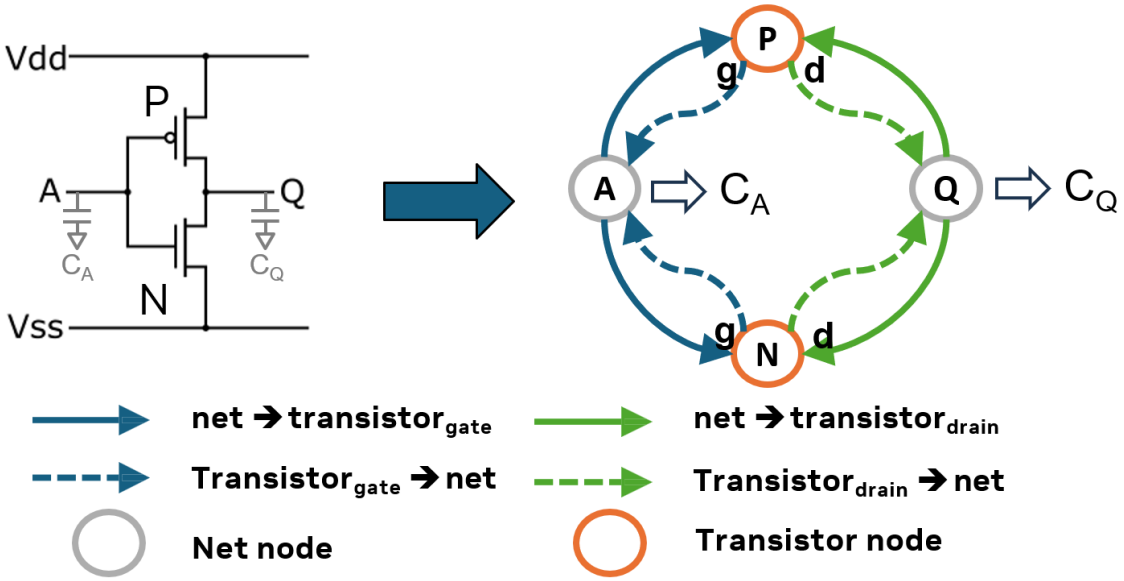
Analysis – Parasitics Prediction

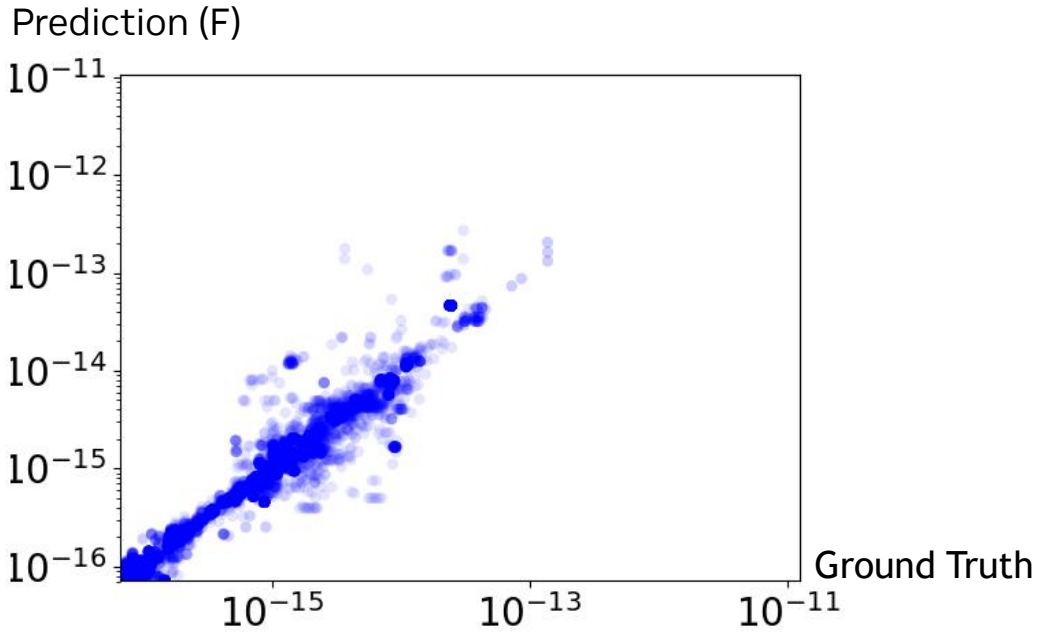
Impact of layout parasitics on schematic design

Use AI to predict layout parasitics from schematic

Convert schematic to graph and learn with GNN



Circuit Schematics to Heterogenous Graph Conversion



MAE=0.852fF MAPE=15%
Simulation error reduced to <10%

H. Ren et al, ParaGraph: Layout Parasitics and Device Parameter Prediction using Graph Neural Networks

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Optimization – Macro Placement

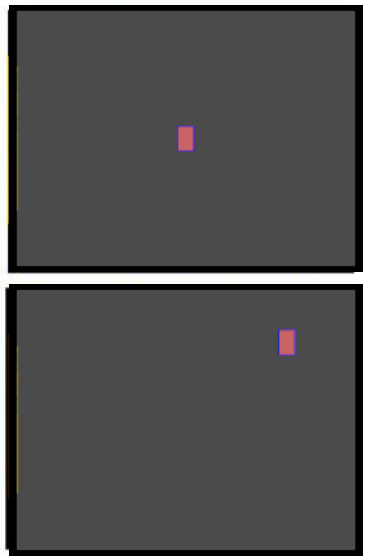
Macro placement quality is very important for physical design

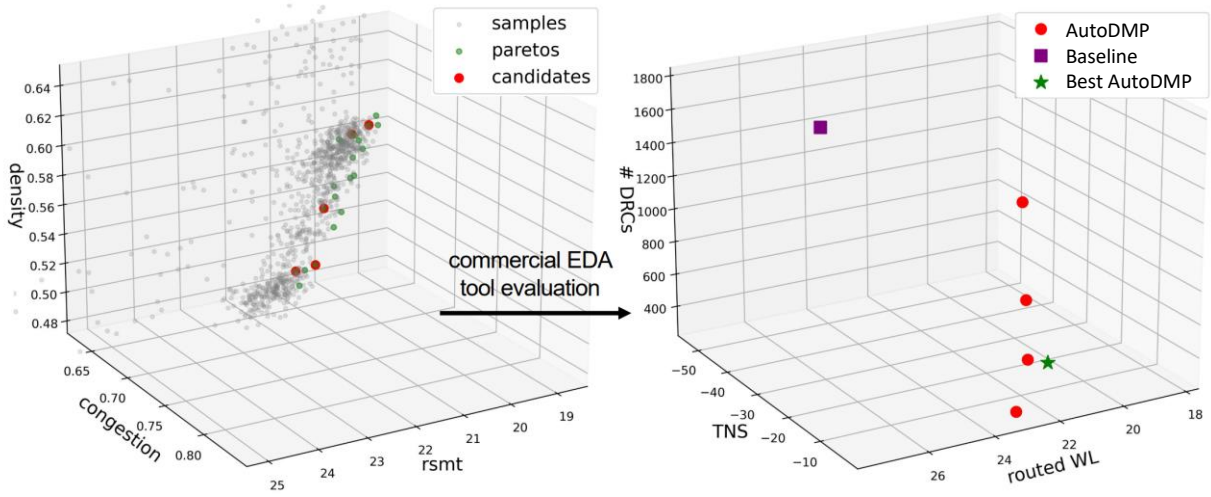
Placement parameters have a huge impact on macro placement

Multi-objective Bayesian optimization: wirelength, congestion, density

Find better macro placement with open-source GPU accelerated placement tools

Parameter	Search Range
*horiz. initial position	[0.2, 0.8] (%)
*vert. initial position	[0.2, 0.8] (%)
*horiz. macro halo	technology dep.
*vert. macro halo	technology dep.
target density d_{target}	$[a_{\text{util}} - 0.2, a_{\text{util}}]$ (%)
density weight	$[1e^{-6}, 1.0]$
smooth HPWL model	{LSE, WA}
smooth HPWL initial γ_0	[0.10, 0.50]
GD initial LR lr_0	$[1e^{-4}, 1e^{-2}]$
GD LR decay	[0.99, 1.0]
GD optimizer	[Adam, Nesterov]
# horiz. global bins	{256, 512, 1024, 2048}
# vert. global bins	{256, 512, 1024, 2048}
λ update lower coeff. L	[0.90, 0.99]
λ update upper coeff. U	[1.01, 1.15]
λ update Δ HPWL_{REF}	$[1.5e^5, 5.5e^5]$





A. Agnesina et al, AutoDMP: Automated DREAMPlace-based Macro Placement

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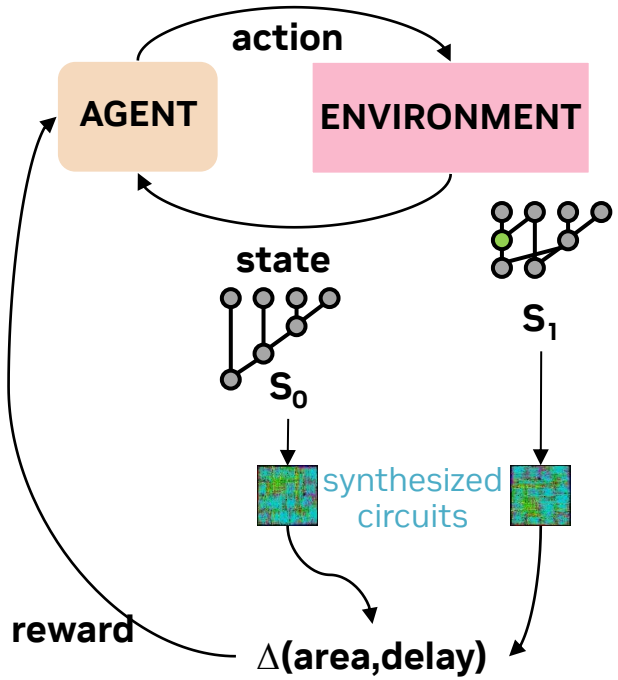
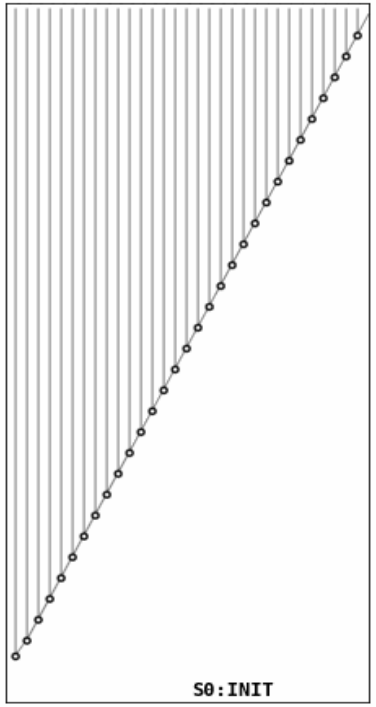
Optimization - Design Better Datapath

Datapath synthesis important for GPU

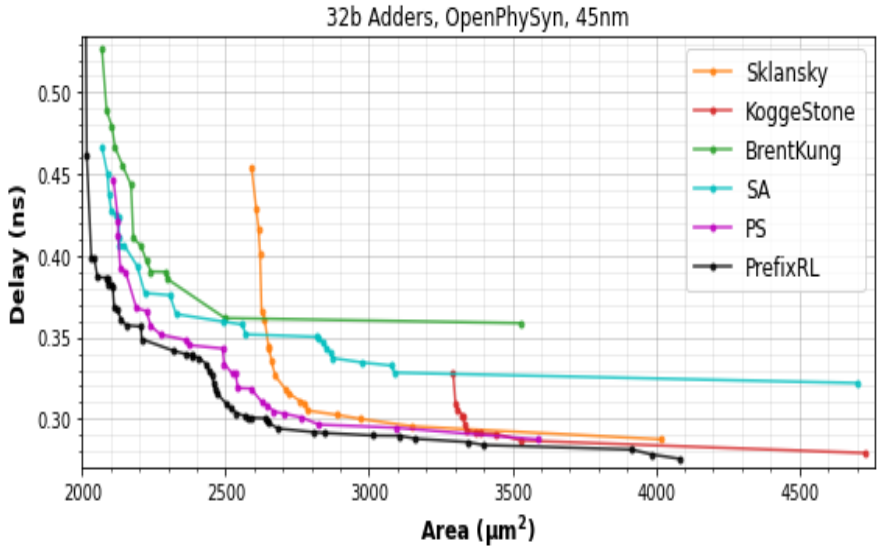
Optimize prefix adder structure with RL



Deep Q learning
Circuit synthesis in-the-loop



Action Space
Add/Delete prefix graph nodes



PrefixRL achieves better results than well known adder architectures

R. Roy et al, PrefixRL: Optimization of parallel prefix circuits using deep reinforcement learning

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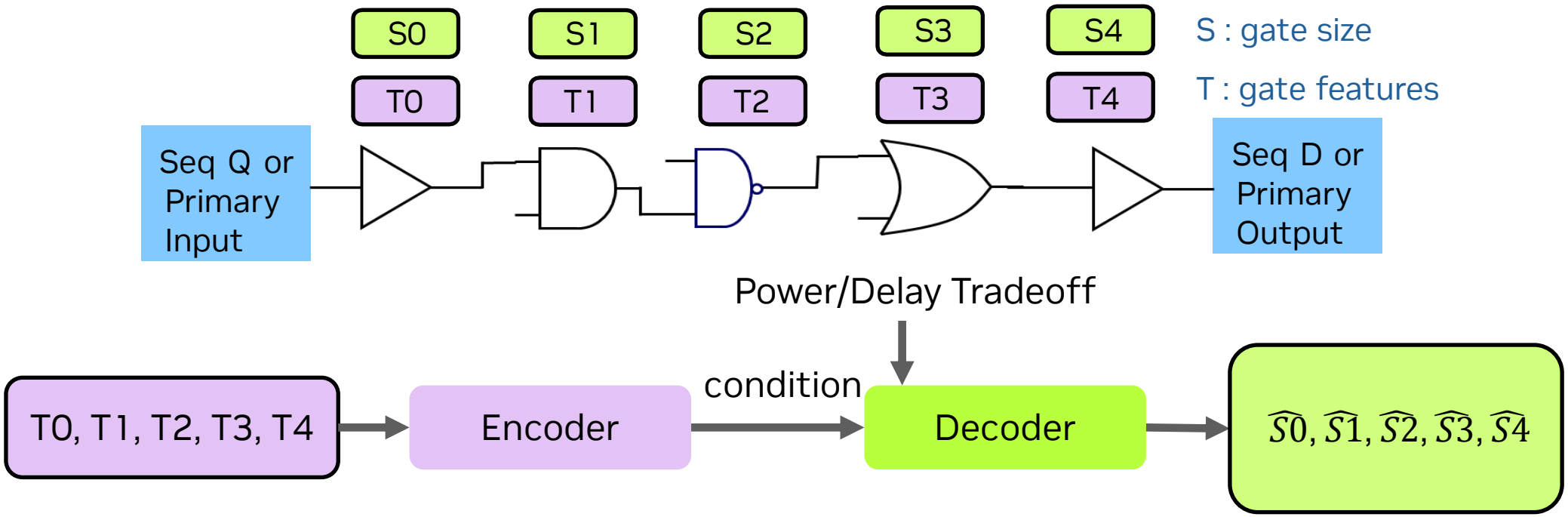
Optimization - Generate Optimal Gate Size

Timing/power optimization such as gate sizing affects scalability of PD tools

Model a path of gates as a sequence, generate optimized gate sizes using Transformer



100X – 1000X speedup compared to traditional optimization with similar PPA

S. Nath et al, Transizer: A Novel Transformer-Based Fast Gate Size

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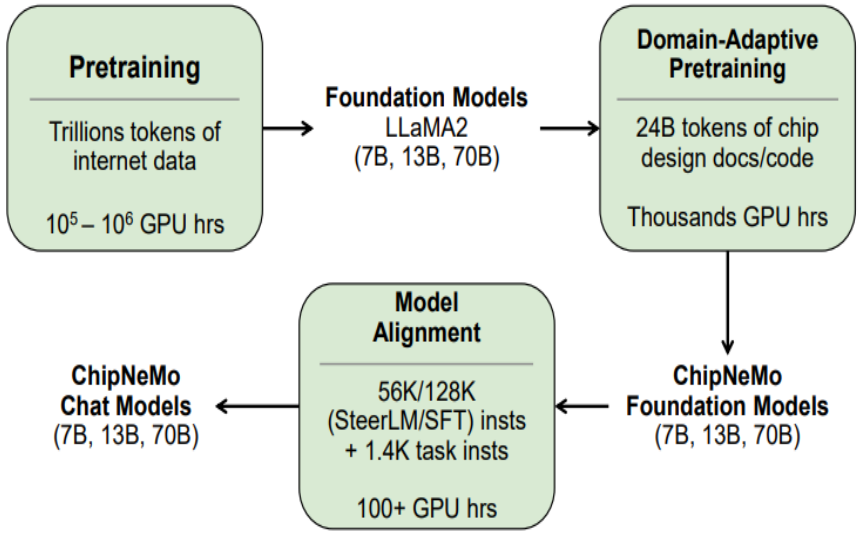
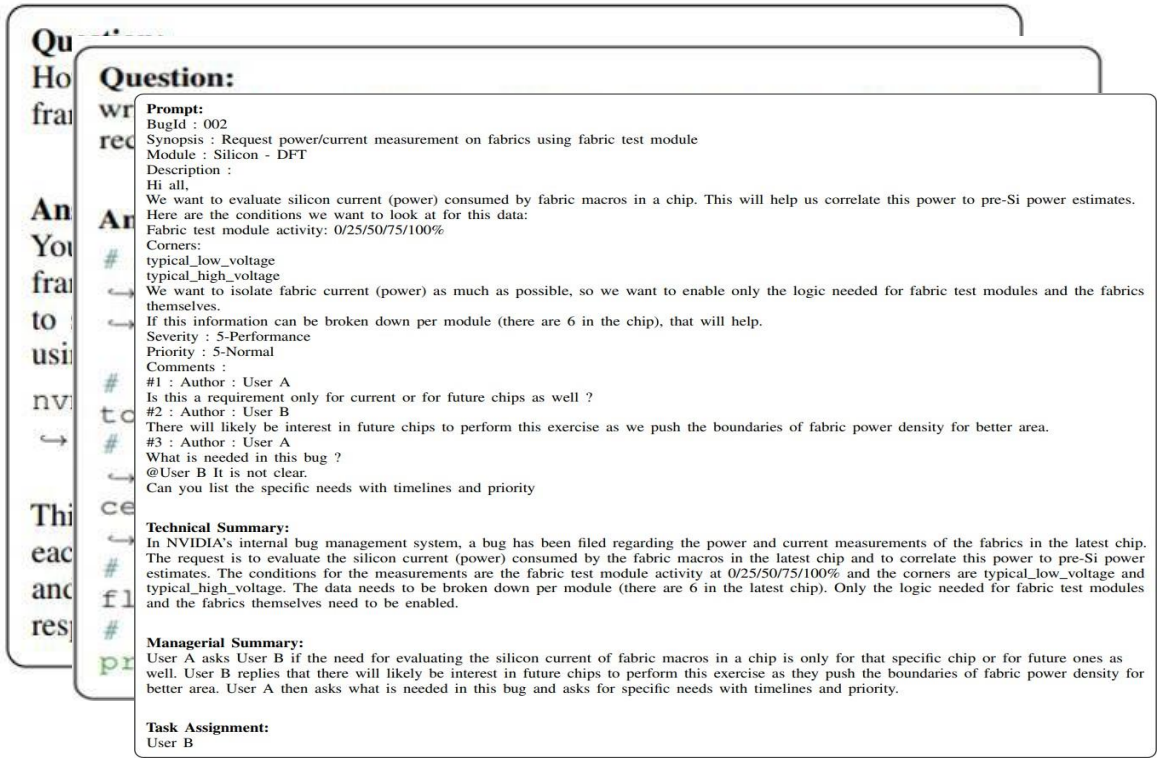
Assistance - ChipNeMo

Answer questions about designs, infrastructures, tools, flows, HW domains, etc.
Generate scripts for specific tasks (VLSI)
Summarize bug reports, predict task assignment





Question:

Prompt:
BugId : 002
Synopsis : Request power/current measurement on fabrics using fabric test module
Module : Silicon - DFT
Description :
Hi all,
We want to evaluate silicon current (power) consumed by fabric macros in a chip. This will help us correlate this power to pre-Si power estimates.
Here are the conditions we want to look at for this data:
Fabric test module activity: 0/25/50/75/100%
Corners:
typical_low_voltage
typical_high_voltage
We want to isolate fabric current (power) as much as possible, so we want to enable only the logic needed for fabric test modules and the fabrics themselves.
If this information can be broken down per module (there are 6 in the chip), that will help.
Severity : 5-Performance
Priority : 5-Normal
Comments :
#1 : Author : User A
Is this a requirement only for current or for future chips as well ?
#2 : Author : User B
There will likely be interest in future chips to perform this exercise as we push the boundaries of fabric power density for better area.
#3 : Author : User A
What is needed in this bug ?
@User B It is not clear.
Can you list the specific needs with timelines and priority

Technical Summary:
In NVIDIA's internal bug management system, a bug has been filed regarding the power and current measurements of the fabrics in the latest chip. The request is to evaluate the silicon current (power) consumed by the fabric macros in the latest chip and to correlate this power to pre-Si power estimates. The conditions for the measurements are the fabric test module activity at 0/25/50/75/100% and the corners are typical_low_voltage and typical_high_voltage. The data needs to be broken down per module (there are 6 in the latest chip). Only the logic needed for fabric test modules and the fabrics themselves need to be enabled.

Managerial Summary:
User A asks User B if the need for evaluating the silicon current of fabric macros in a chip is only for that specific chip or for future ones as well. User B replies that there will likely be interest in future chips to perform this exercise as they push the boundaries of fabric power density for better area. User A then asks what is needed in this bug and asks for specific needs with timelines and priority.

Task Assignment:
User B

M. Liu et al, ChipNeMo: Domain-Adapted LLMs for Chip Design

Challenges

Analysis

“Inaccurate predictions”

Optimization:

“AI is not as good as existing algorithms”

“AI will never get better results than the data it trained on”

Assistance

“Hallucinations”

“Can not solve complex problems”

Compare AI and Algorithm

	Algorithm	AI
Categories	Placement, route, synthesis, CTS, etc	Supervised, unsupervised, reinforcement learnings, etc
Optimality	Known	Unknown
Robustness	Works on any data distribution	Do not work if training and inference distribution mismatch
Training	No	Require a lot data
Interpretability	Behavior explainable	Not explainable
Pros	Solve a known problem efficiently	Solve any problem by learning from complex data
Cons	Oversimplification of dynamic, complex problem	Difficult to leverage the mechanics of the problem

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Similar Discussions

Two paradigms for Artificial Intelligence

The logic-inspired approach

The essence of intelligence is using symbolic rules to manipulate symbolic expressions.

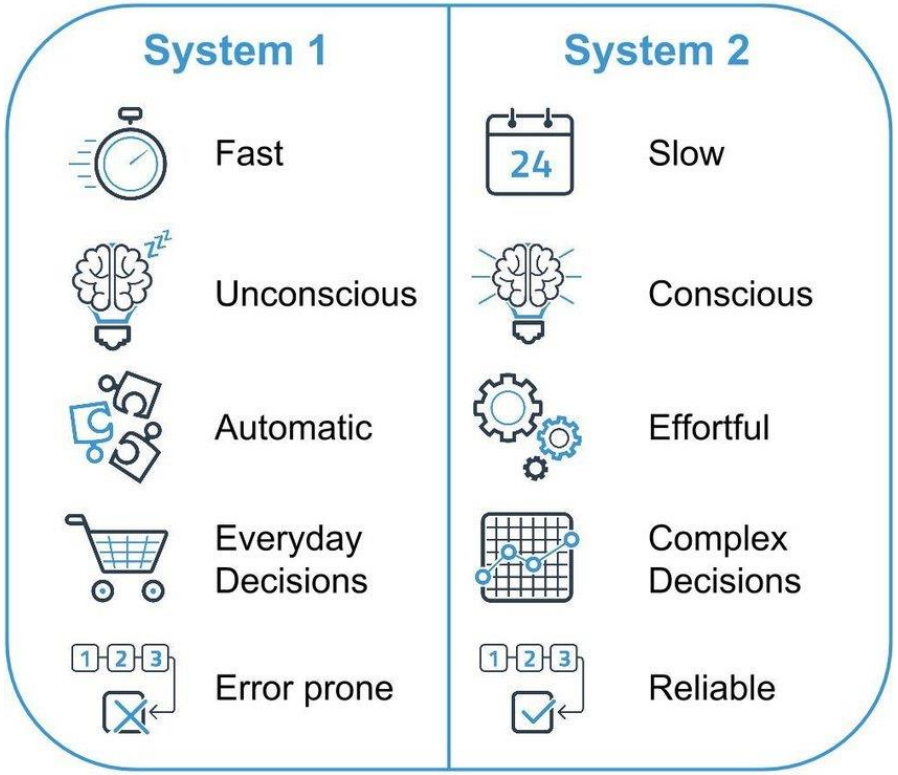
We should focus on reasoning.

The biologically-inspired approach

The essence of intelligence is learning the strengths of the connections in a neural network.

We should focus on learning and perception.

Jeff Hinton: Turing Award Lecture

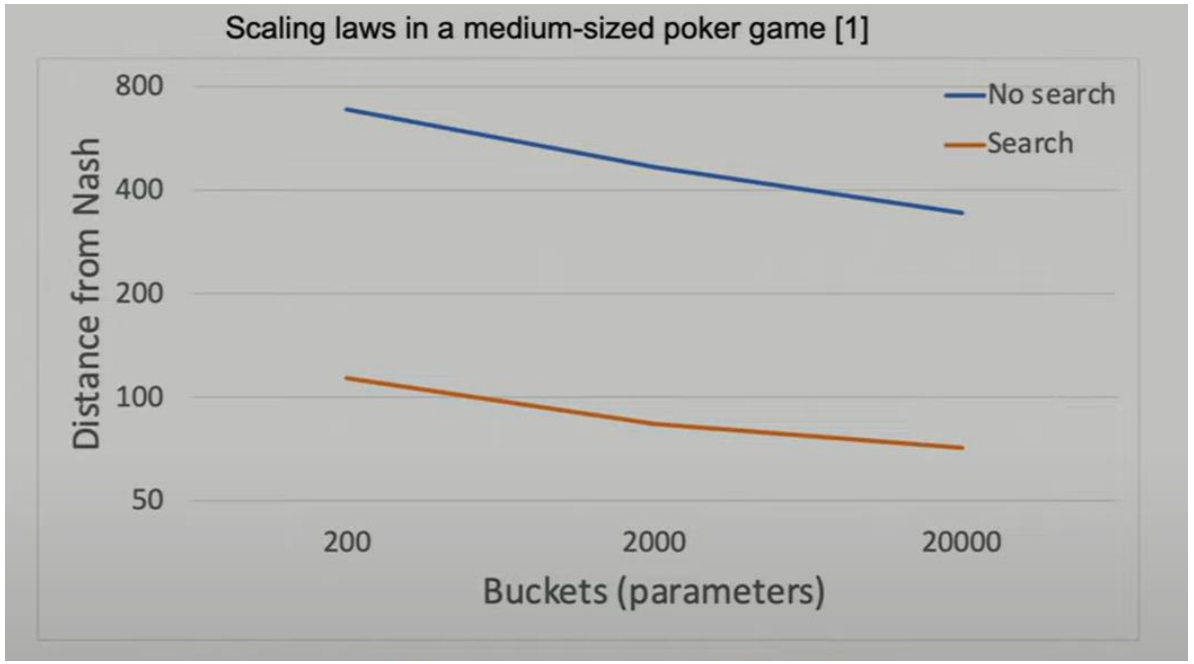


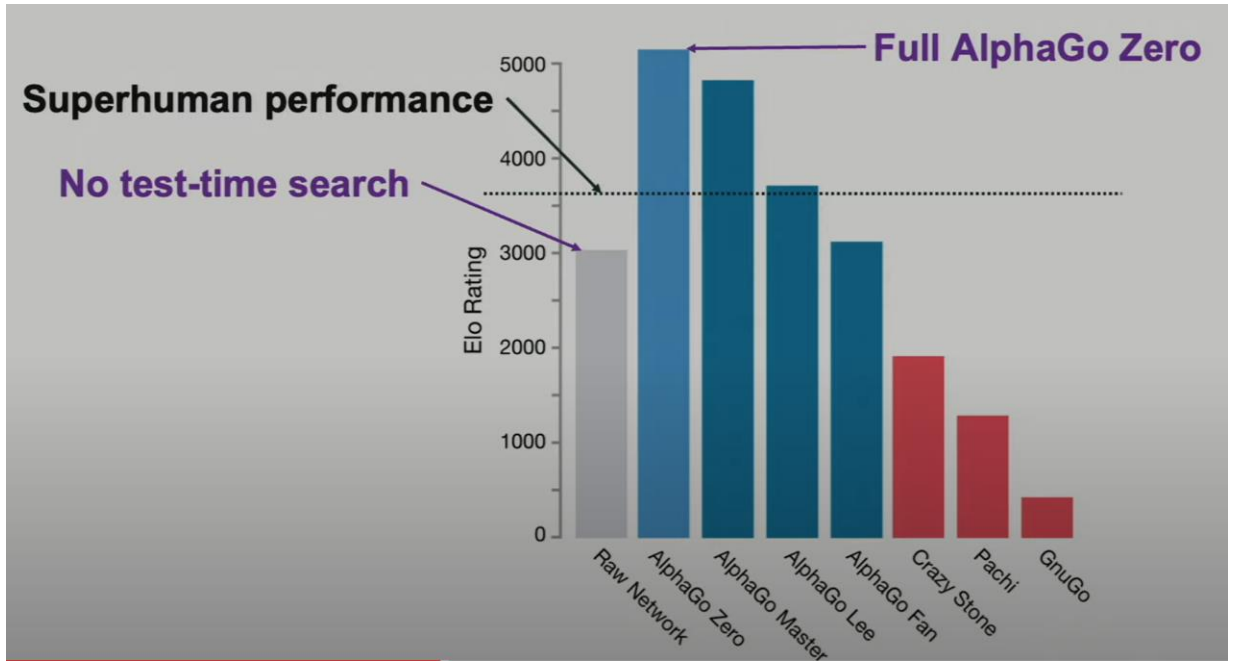
Daniel Kahneman: Thinking, Fast and Slow

Integrate AI with the Algorithm @ Inference

Example – Gaming

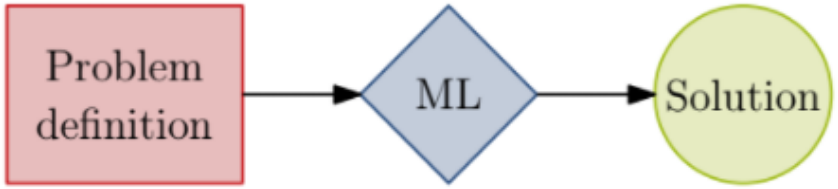
Search helps AI game model perform much better



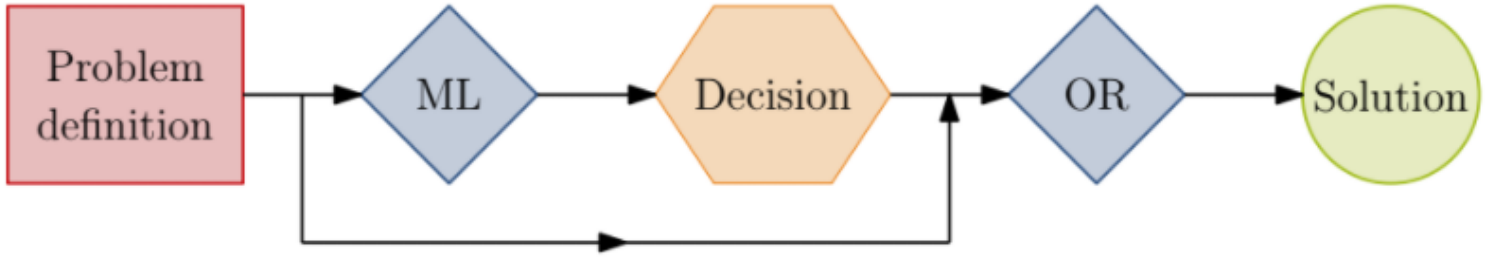


Noam Brown: Parables on the Power of Planning in AI: From Poker to Diplomacy

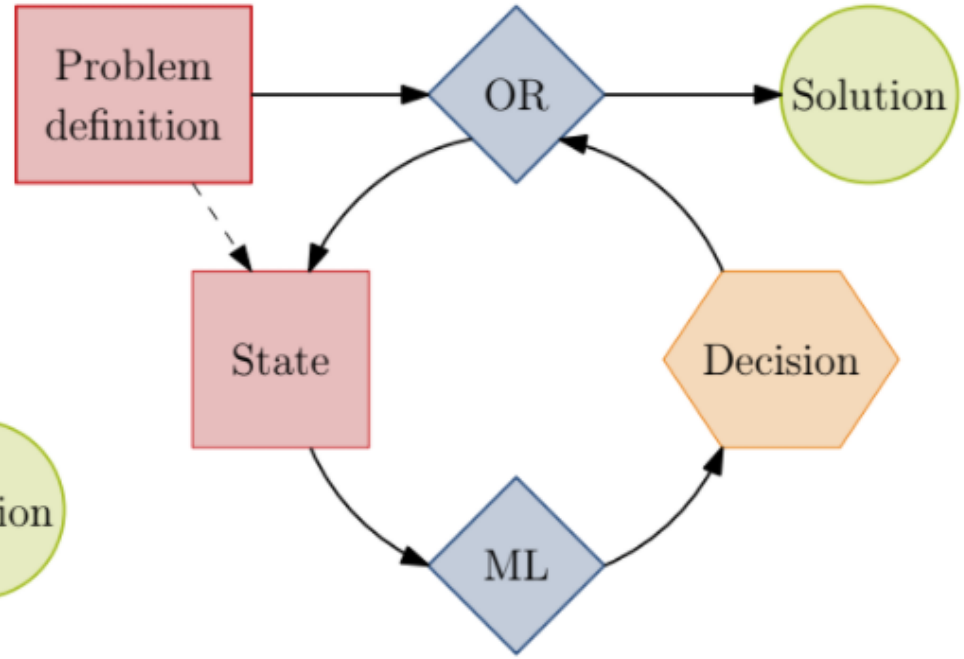
ML for Combinatorial Optimization



End-to-end learning



Coarse grain integration



Fine grain integration

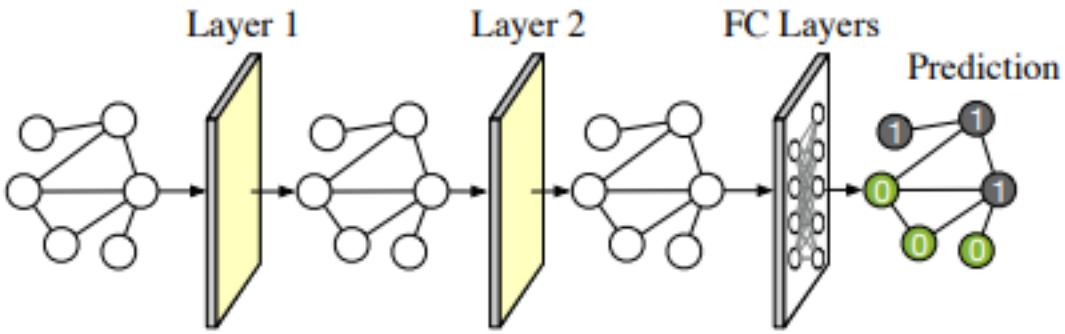
Yoshua Bengio et al, Machine Learning for Combinatorial Optimization: a Methodological Tour d'Horizon

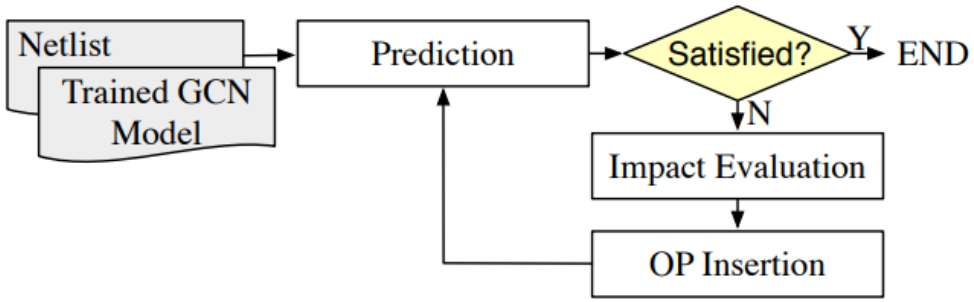
TPI: AI as Oracle for Algorithm

GCN model predicts node testability: Difficult-to-test nodes (DTN) prediction

Model acts as a predictor/oracle for a test point insertion (TPI) algorithm

Although model accuracy is only 90%, reduced test points by 11% and test patterns by 6% to achieve similar coverage as a commercial test point insertion algorithm





Y. Ma et al, High Performance Graph Convolutional Networks with Applications in Testability Analysis

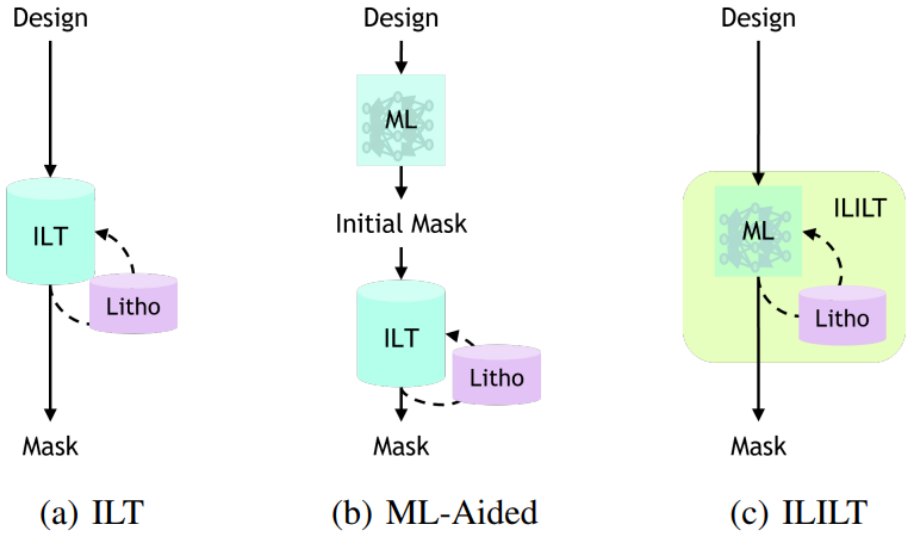
ILILT : Ground AI with Algorithm

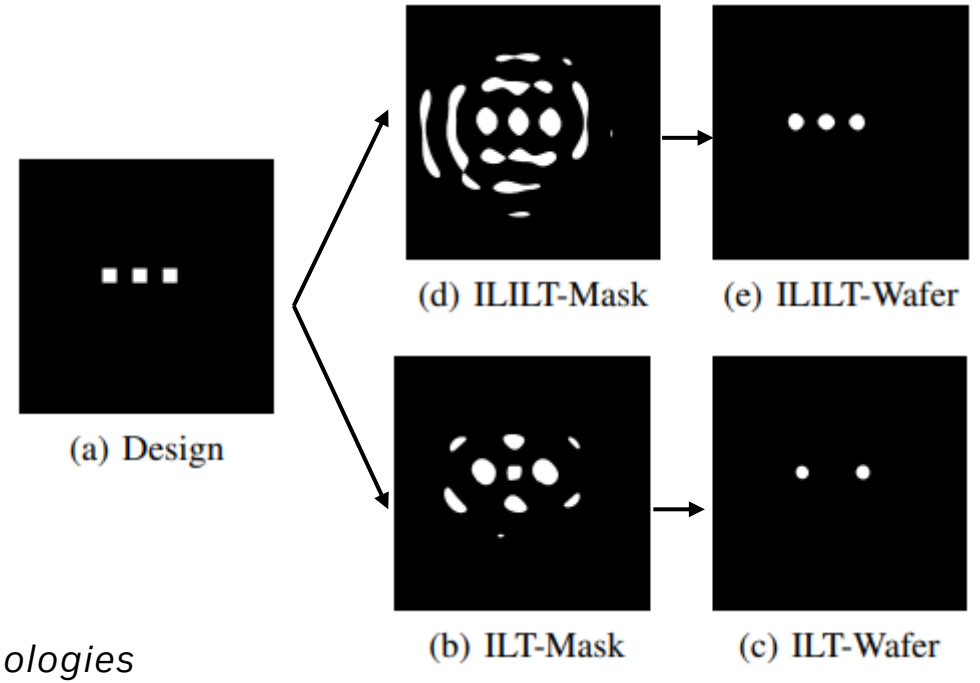
Learn the optimized mask from ILT (inverse lithography Technology) solver

Use Litho forward simulation algorithm to ground the model

Increase inference time compute with multi-step inference

Better quality than ILT (generated the training data) with 10X speedup





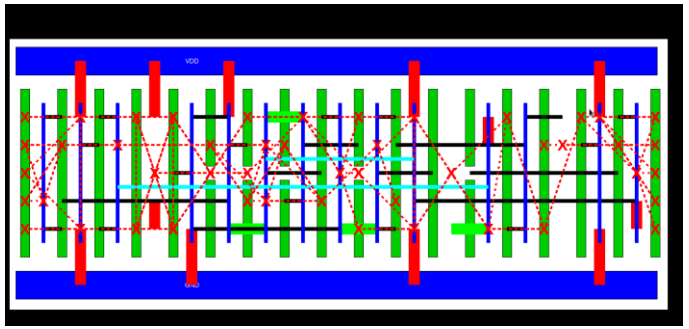
H. Yang et al, ILILT: Implicit Learning of Inverse Lithography Technologies

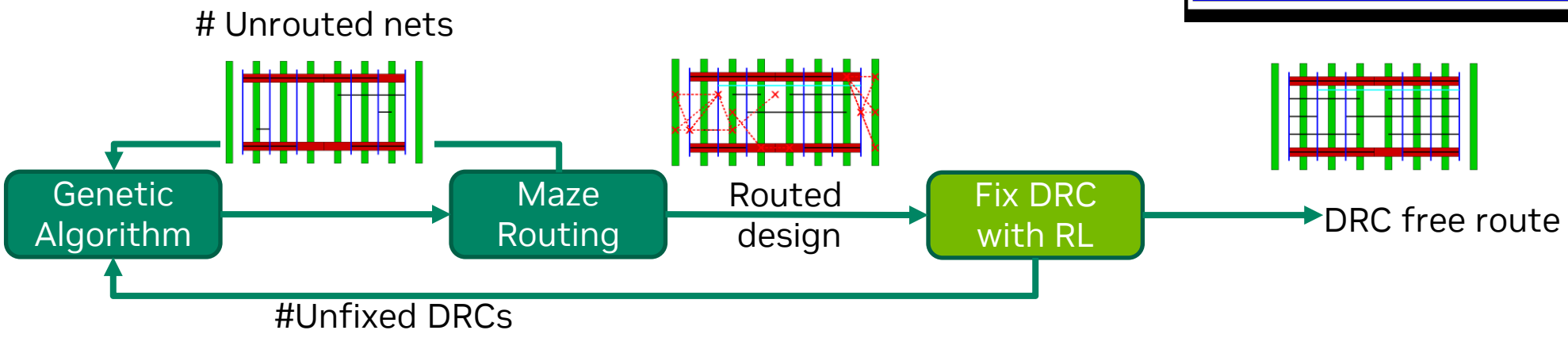
NVCell: AI as Part of the Algorithm

Auto routing for std cells, overcoming DRC challenge

Use an algorithm to route, but RL to fix DRC

Generate std cell layout of entire industrial libraries with better quality than designer





H. Ren et al, Standard Cell Routing with Reinforcement Learning and Genetic Algorithm in Advanced Technology Nodes

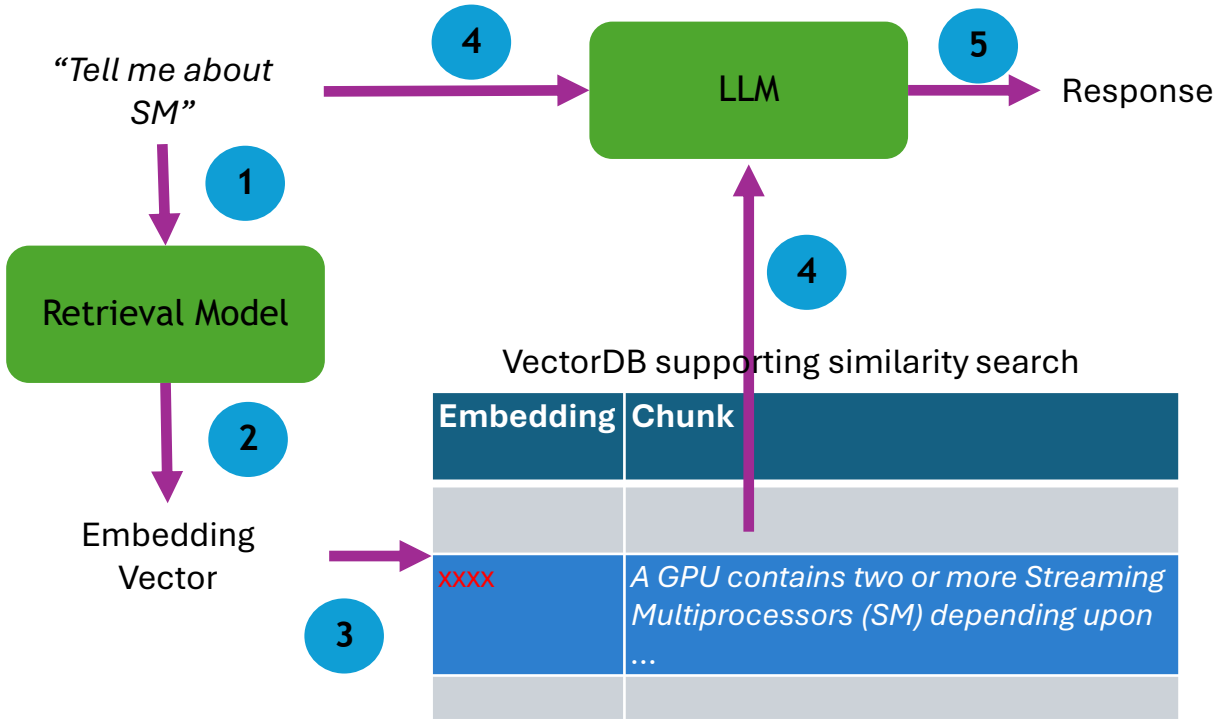
RAG : Algorithm @ Inference

Retrieval Augmented Generation (RAG) helps to ground LLM with curated documents

Help reduce hallucination

- Incorrect, outdated, conflicting, scarcity of the training data

RAG leverages the search algorithm (embedding similarity “Hash”) to retrieve the relevant chunks as context for the question

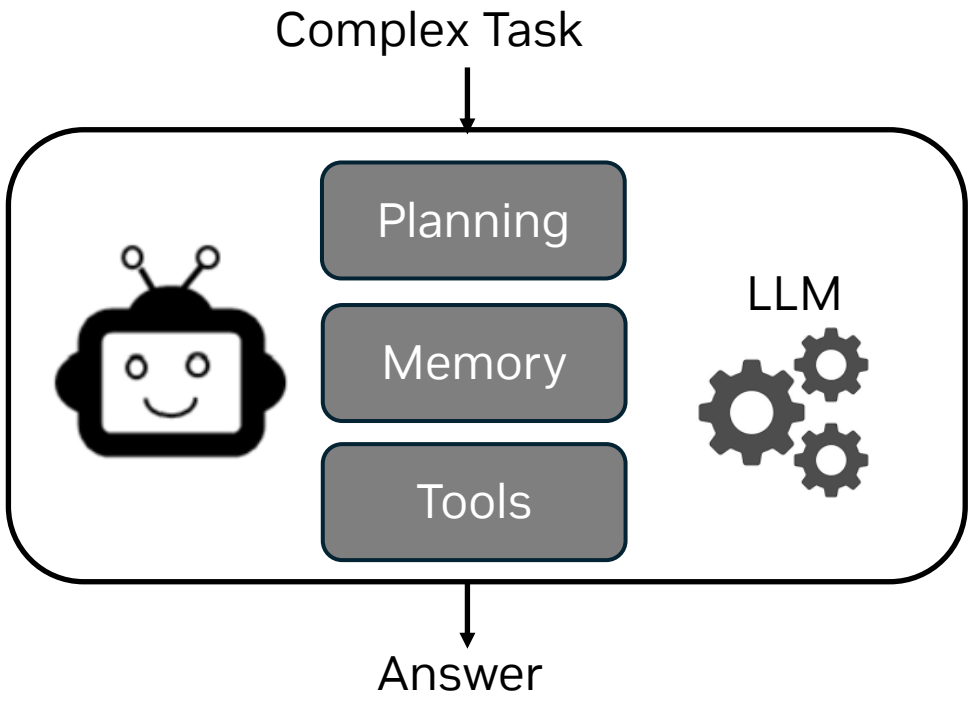


RAG Illustration

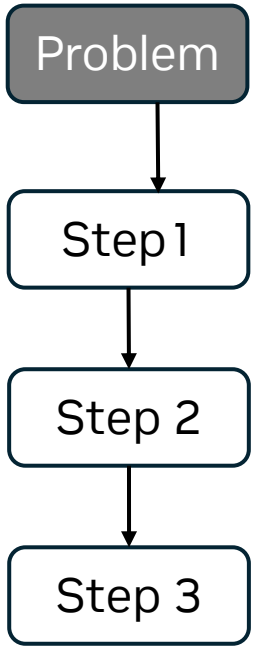
Agentic System: Algorithm @Inference

Problem decomposition enables Agent to handle complex tasks

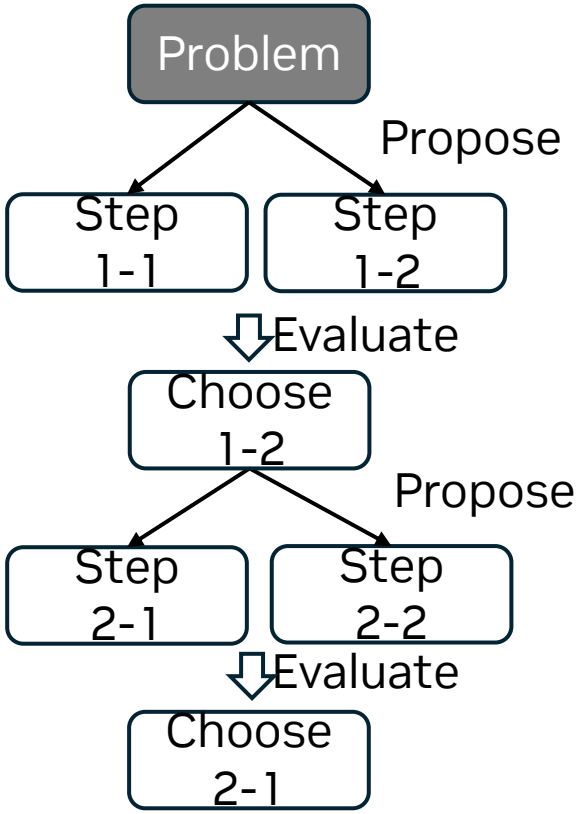
Agent uses algorithms with LLM as compute unit



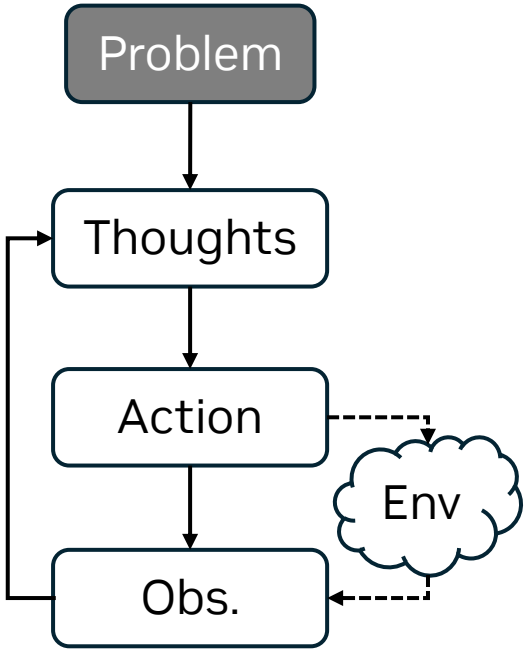
Planning



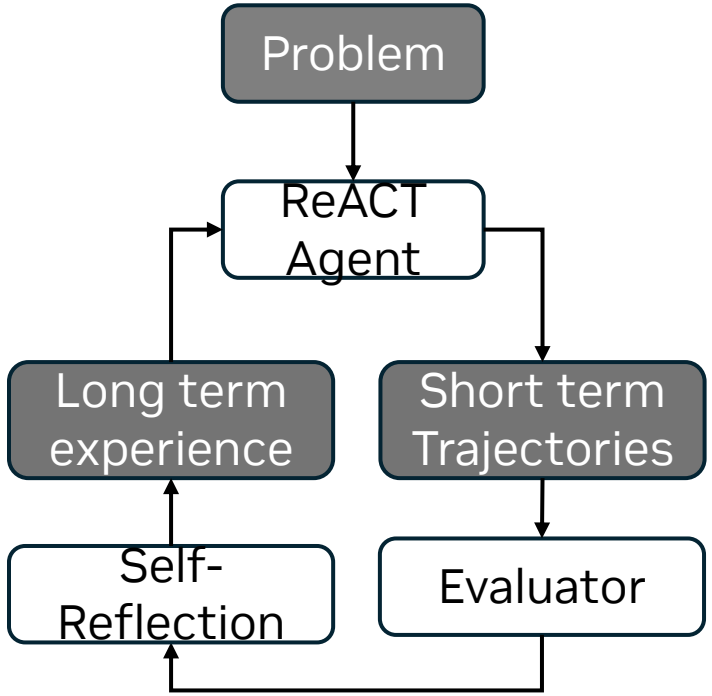
Chain-of-Thought
(Procedure)



Tree-of-Thought
(Branch)



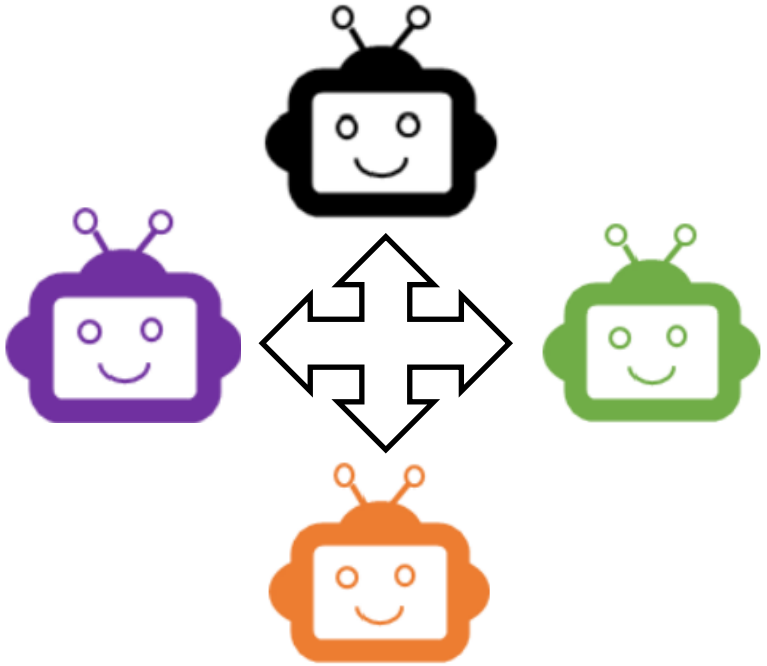
ReACT
(Loop)



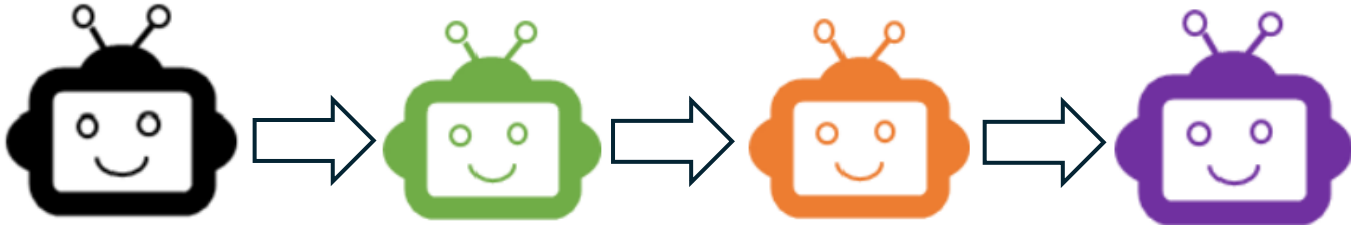
Self-Reflection
(Multi-Level Loop)

Multi-Agent

Hierarchical decomposition with dynamic and static structures



Agent Conversation
(dynamic)



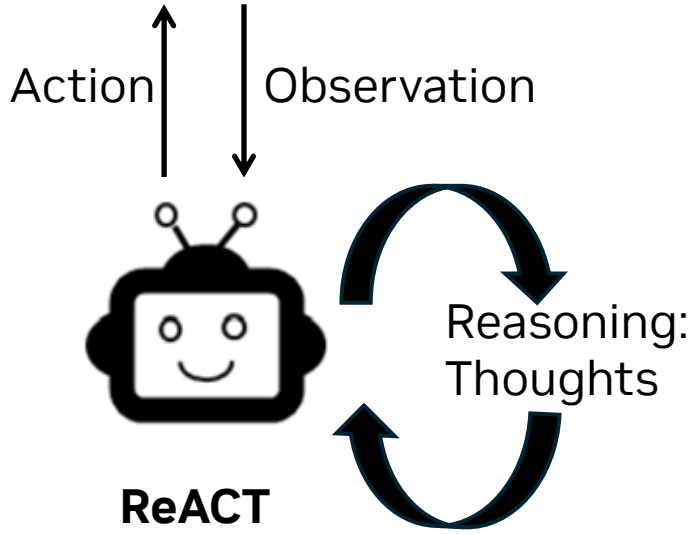
Task Flow
(static)

Timing Report Analysis

Tools

timing_metric_calculation_tool :
Calculate changes in WNS, TNS, or FEP

slack_distribution_calculation_tool:
Calculate changes in the slack distribution



ReACT

CoT prompting

Let's think step by step.

Firstly, analyze the change of WNS, and TNS for all designs in all PVT corners tables of these two "datecode" settings and explain with numbers using tools.

Secondly, comparing "FEP" of two datecode settings in all PVT corners tables.

Thirdly, analyze the slack distribution to identify the distribution of "slack less than 0" paths.

Finally, summarize the analysis in the following aspects:

1. *Provide key takeaways, comparison, and suggestions with bullet points of the two "datecode" settings based on previous steps.*
2. *Identify the corner which still suffers many timing violations if any.*

You need to use the provided tools to analyze the timing metrics! You are not good at math!

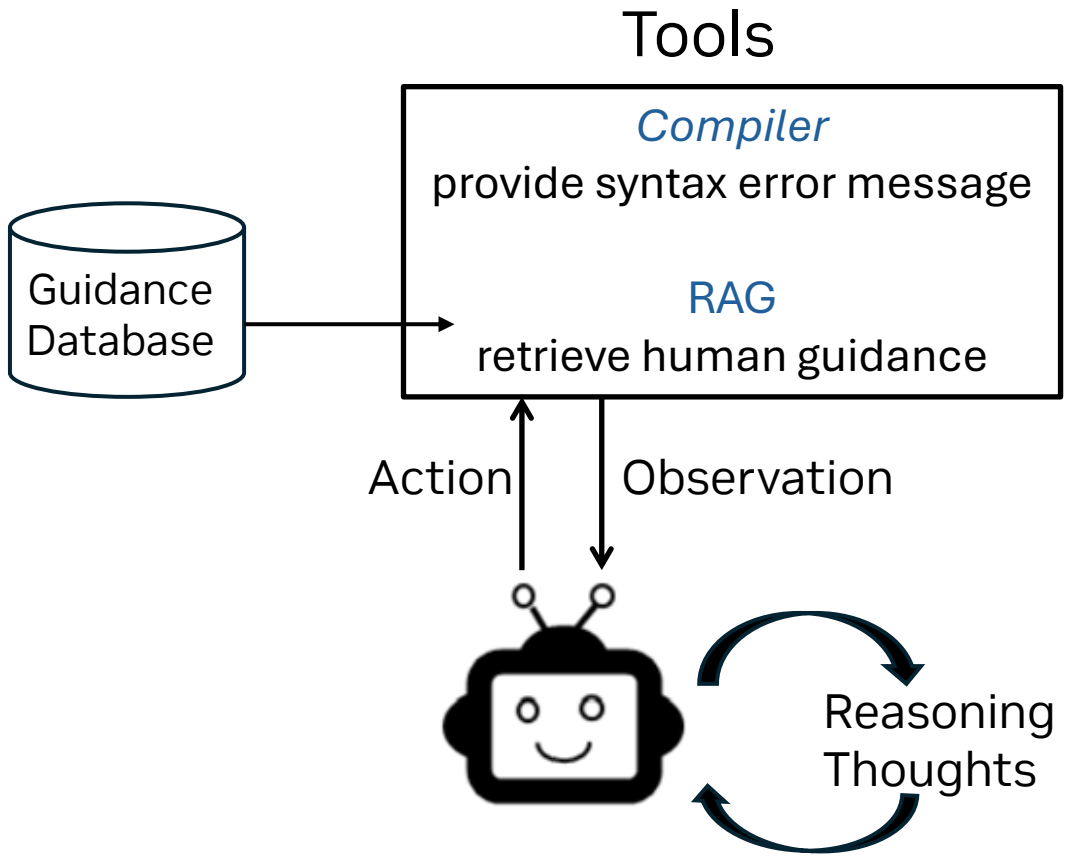
RTLFixer : Fix Syntax Error in RTL code

55% of GPT-3.5 Verilog generation errors are Syntax errors

Leverage **ReAct** framework to drive Verilog compiler to get feedback

Leverage **RAG** to retrieve human guidance for each syntax error

Fixing 99% of Syntax Errors on VerilogEval-syntax Benchmark



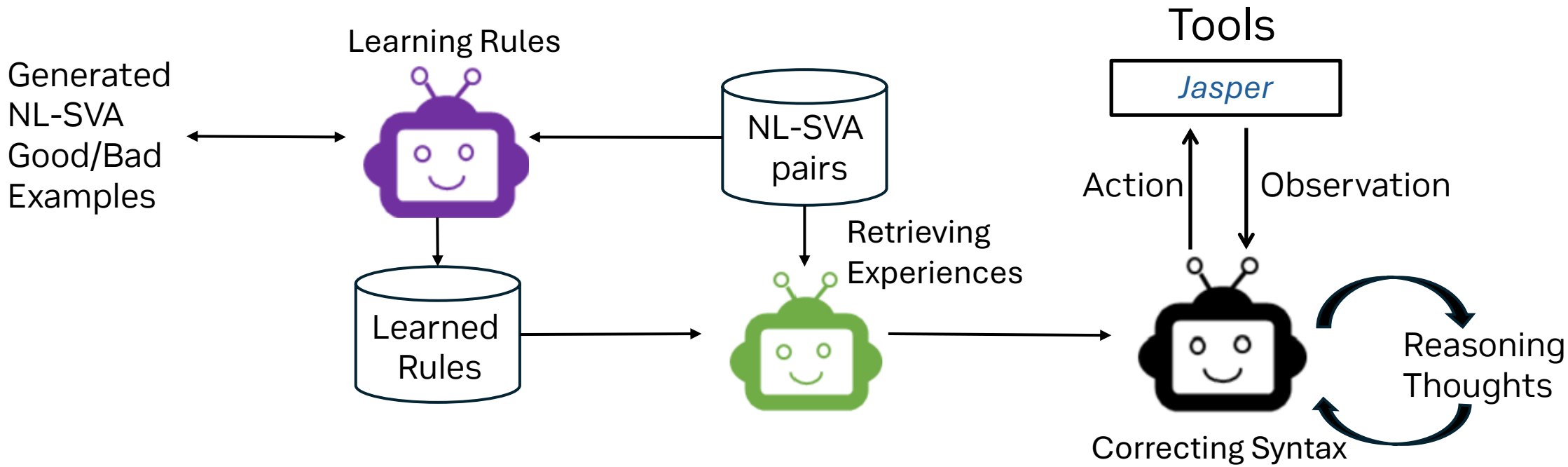
Y.-D. Liu et al, RTLFixer: Automatically Fixing RTL Syntax Errors with Large Language Models

FVAgent : Generate SVA From NL

Self-Learning: Generate rule knowledge base for RAG

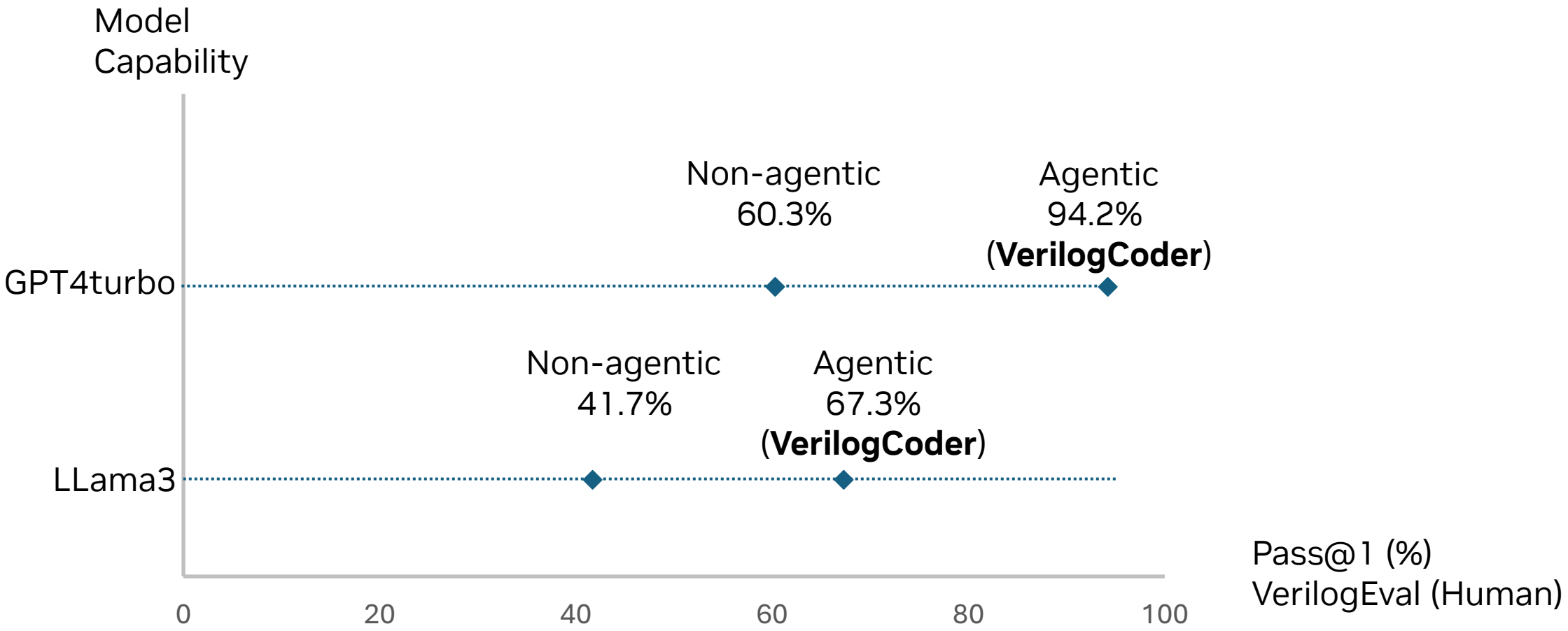
Multi-Agent task flow : Retrieval: prior NL-SVA pairs as well as learned rules; Correct syntax errors with tools

Improve Syntax/Function corrections on all models



J. Wan et al, FVAgent: Bridging Natural Language Specifications and Formal Verification Assertions with Adaptive Multi-Agent Learning

VerilogCoder Agent



C.-T. Ho et al, VerilogCoder: Autonomous Verilog Coding Agents with Graph-based Planning and Abstract Syntax Tree (AST)-based Waveform Tracing Tool

VerilogCoder Agent

Dynamic task planning and execution

Tasks executed by single or multi-agents

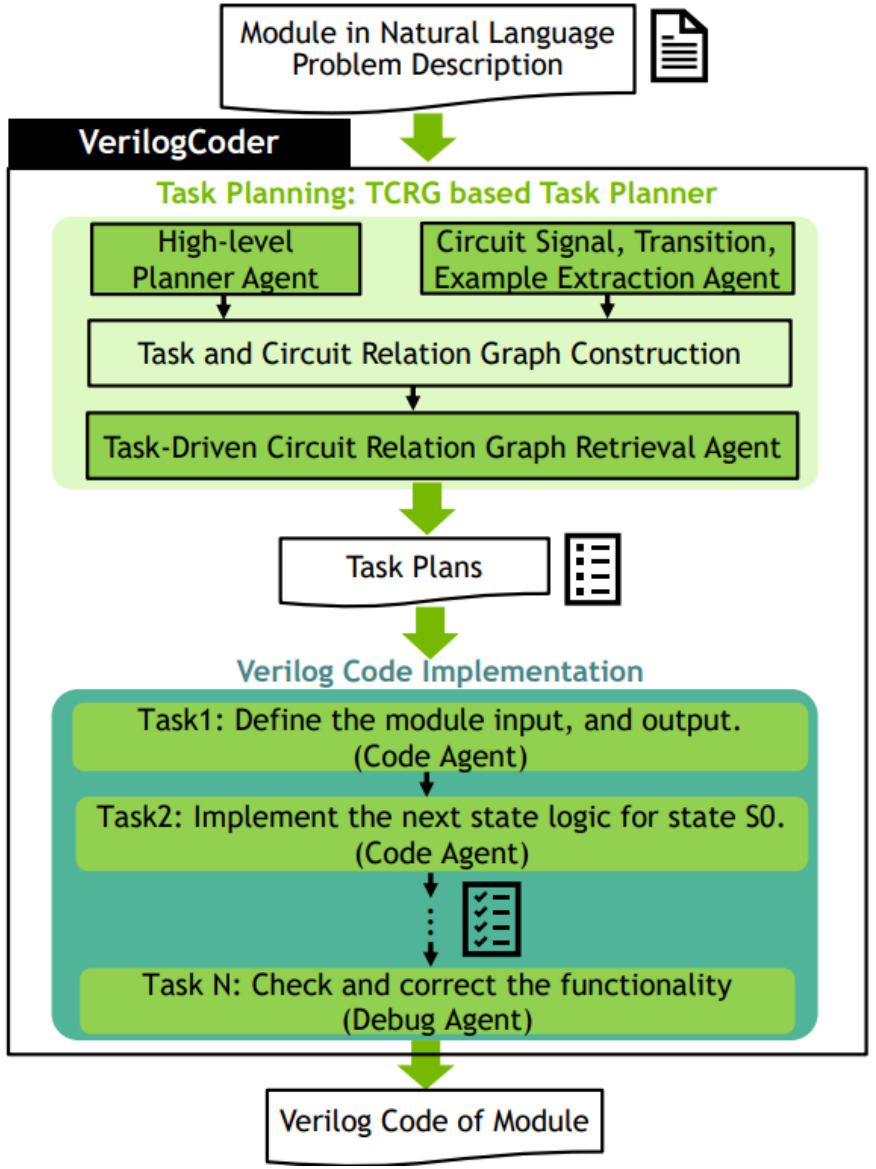
Special knowledge base and tools

Task Planning

Task-Driven Circuit Relation Graph (TCRG)

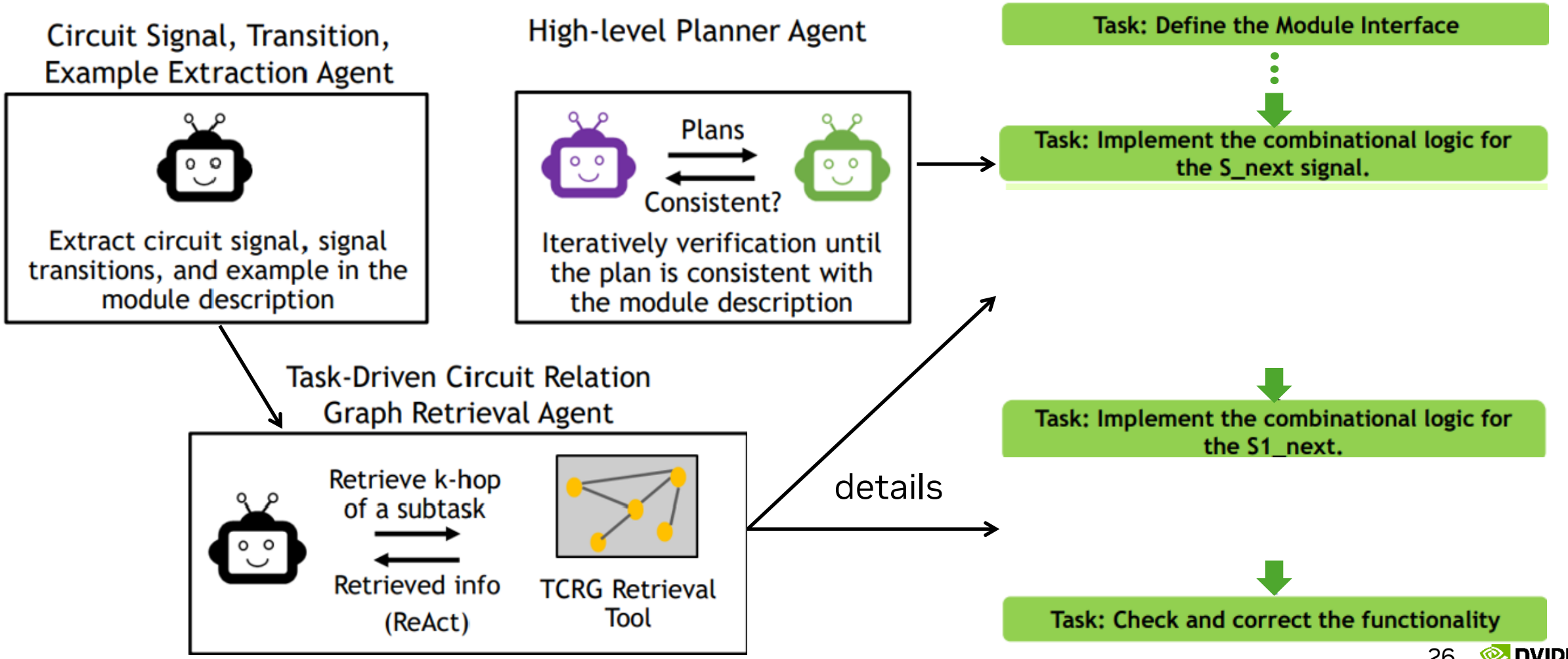
Code Implementation

AST-guided waveform debugging tool



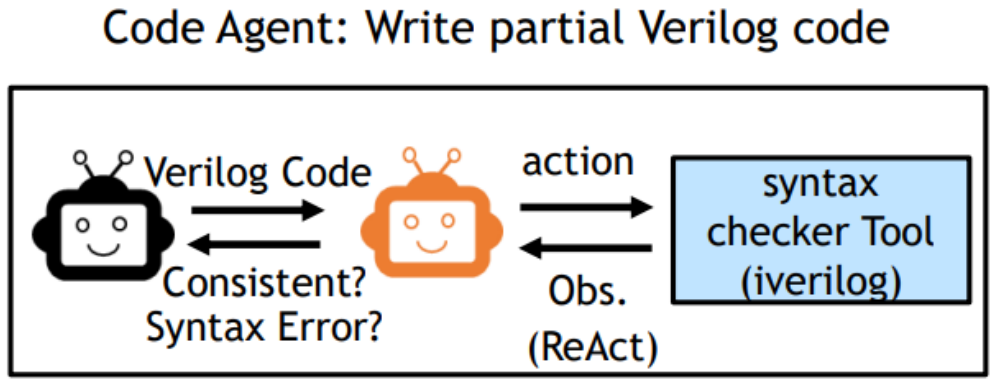
C.-T. Ho et al, VerilogCoder: Autonomous Verilog Coding Agents with Graph-based Planning and Abstract Syntax Tree (AST)-based Waveform Tracing Tool

Task Planning

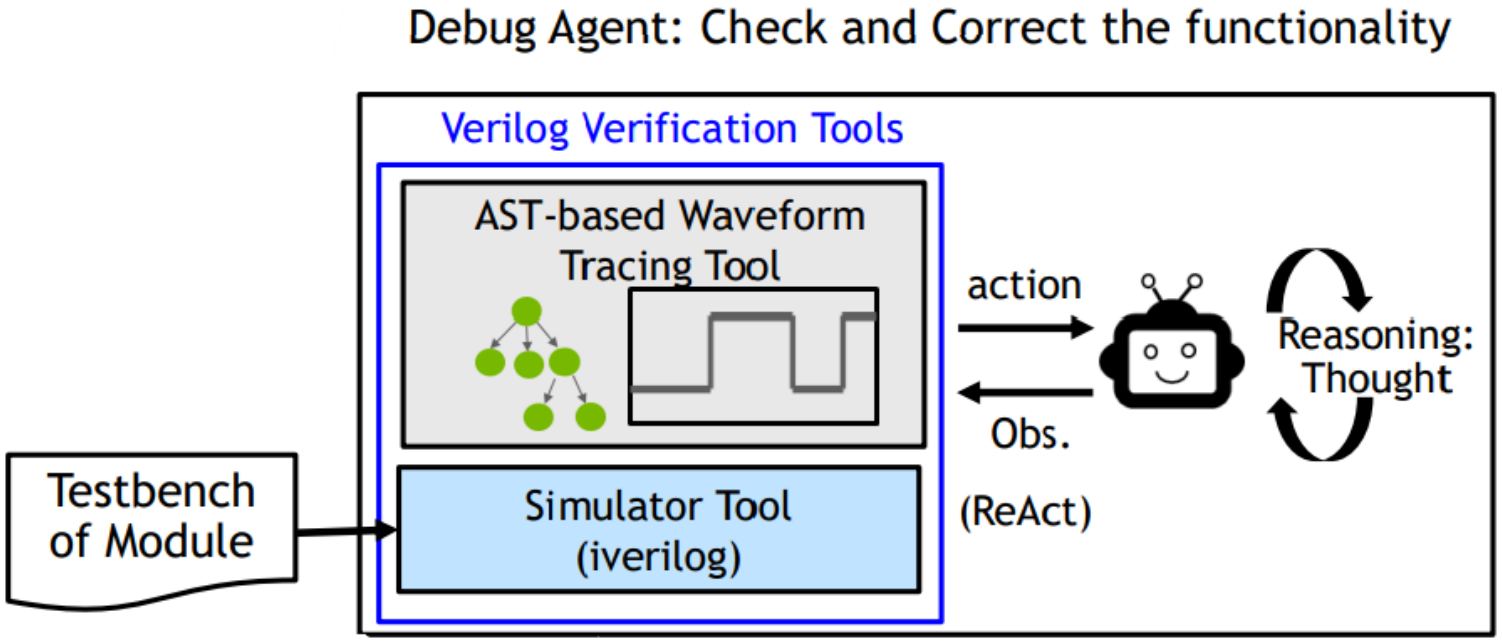


Code Implementation

Writing code for each task



Debug final code



Future – AI Learning the Algorithm

Learning to decompose a problem into steps and beyond?

Write a bash script that takes a matrix represented as a string with format '[1,2],[3,4],[5,6]' and prints the transpose in the same format.

Formulating the problem

Crafting the script

Reading input and parsing

Breaking down the matrix

Determining matrix dimensions

Mapping out the matrix

Constructing the output

bash Copy code

```bash
#!/bin/bash

INPUT="$1"

# Remove '[' and ']' and replace '],[' with newline
rows=$(echo "$INPUT" | sed 's/\],\[/\n/g' | sed 's/\[//g' | sed 's/\]//g')

# Read lines into array
mapfile -t lines <<< "$rows"

nrows=${#lines[@]}
# Get number of columns from the first line
cols=$(echo "${lines[0]}" | awk -F',' '{print NF}')

declare -a matrix

# Build the matrix
for ((i=0; i<nrows; i++)); do
    IFS=',' read -ra elements <<< "${lines[i]}"
    for ((j=0; j<${#elements[@]}; j++)); do
        index=$((i * cols + j))
        matrix[$index]=${elements[j]}
    done
done
```

"It learns to break down tricky steps into simpler ones. It learns to try a different approach when the current one isn't working. This process dramatically improves the model's ability to reason."

OpenAI, Learning to Reason with LLMs

NVIDIA

Closing Thoughts

Meet the challenges of AI systems with deep integration of AI and Algorithms

Leverage the advantages and avoid the pitfalls of either one

Explore the vast potential of integrating AI with algorithms for EDA problems: design analysis and optimization.

Exploit the revolutionary advancement of the Agentic system (an algorithm using LLM as a computing node) for design assistance

Algorithms designed by human based on their domain knowledge; but could be learned by AI in the future.

NVIDIA